Sergio Toro Codo

Robotics & Automation Engineering Student - Embedded Systems, Computer Vision, Robotics

storocodo@gmail.com | Nanjing, CN / Zaragoza, ES

PROFESSIONAL SUMMARY

Electronics and Automation Engineering student (Universidad de Zaragoza, EINA), on an exchange year in Mechanical Engineering at Nanjing Tech University (China). 3+ years building autonomous mobile robots end-to-end: mechanical design, custom PCB electronics, embedded firmware and control software. IEEE student branch leader (robotics & rocketry). Strong hands-on profile in robotics, embedded systems, computer vision, deep learning, control theory (PID), sensor fusion, SLAM/odometry, simulation and testing, with industrial internship experience. Interested in autonomous/industrial robotics, mapping & localization, simulation and hardware testing.

TECHNICAL SKILLS

- **Programming:**Python (Tkinter, NumPy, OpenCV, Plotly), C/C++ (embedded), MATLAB, SQL, Git/GitHub, Linux, PlatformIO, Arduino.
- **Embedded & Electronics:**ESP32/ESP32-S3/ESP32-CAM, PCB design (KiCad, Altium), power electronics (DC-DC MP1584EN), motor drivers (H-bridge), sensors: ToF (VL53L0X/L1X), ultrasonic, IMU (BNO055), encoders; buses I2C (XSHUT multiplexing), SPI, UART; ESP-NOW, Bluetooth, Wi-Fi, LoRa telemetry.
- **Control & Robotics:**PID/PI control with anti-windup, odometry / differential kinematics, finite state machines (FSM), hierarchical control, SLAM fundamentals, TinyML edge inference, LIDAR-based localization.
- **Computer Vision & ML:**OpenCV, PyTorch, TensorFlow/Keras, SIFT/ORB, SuperPoint/SuperGlue, RANSAC, homography, fundamental/essential matrices, DLT triangulation, PnP, optical flow, incremental SLAM, transfer learning (MobileNetV3, EfficientNet, Xception), CNNs, FCN.
- **CAD & Manufacturing:**SolidWorks, Fusion 360, 3D printing (FDM), technical drawing, mechanical integration.
- **Languages:** Spanish (native), English (B2), Mandarin Chinese (HSK 2, ongoing).

PROFESSIONAL EXPERIENCE

Board Member - Electronic Rocketry & Robotics Section Jan 2026 - Present

IEEE Student Branch UZ, Zaragoza

- **Alunizar (rocketry):**electronic design for in-flight data acquisition and a solid-motor test bench (ESP32-S3, thermocouple MAX31865, 24-bit ADC ADS1220, LoRa E220 telemetry).
- **Robotics Section:** 10th ASTI Robotics Challenge; mentoring students, transferring hands-on embedded & design skills.

R&D Electronic Engineer - Curricular Internship (Robotic Systems) Jun - Aug 2025

Escribano Mechanical & Engineering (EM&E Group / AUNAV), Binéfar, Spain

- PCB design in Altium, PCB error correction, power-supply design, technical research for robotic systems.
- Assembly, testing and validation of robotic systems in an industrial R&D team.

Student Member & Vocal (Marketing & Communications) Oct 2024 - Feb 2026

IEEE SB UZ, Zaragoza

- Branch communications and outreach; student branch activity management.

TECHNICAL PROJECTS

JRG-25 - Autonomous Mobile Robot, ASTI Robotics Challenge (ARC 2026) Dec 2024 - 2026

Team "Mesa Paso 43" (IEEE SB UZ)

Skills developed: mechanical design (SolidWorks/Fusion 360, 3D printing), custom PCB design (KiCad), sensor integration (ToF, IMU, encoders, I2C multiplexing), embedded C++ control (PID with anti-windup, differential odometry, FSM design), layered firmware architecture with CI, GUI development (Python/Tkinter), wireless protocols (ESP-NOW, Bluetooth SPP), project & budget management.

S.P.I.T. - Early Wildfire Prevention System, WED 2026 Hackathon Mar 2026

Team InnoviZar

Skills developed: UAV-based sensing (RGB + thermal imaging), on-board TinyML inference, PLC control with pressure/flow feedback, hydraulic system sizing, cross-disciplinary rapid prototyping under hackathon constraints.

AGROPLANT - NASA Space Apps Challenge 2025 Oct 2025

Hybrid physical-digital agricultural game

Skills developed: ESP32/RFID hardware-software integration, Wi-Fi communication protocols, data-driven design using open scientific datasets.

Repair Café & Workshop (EINA Agenda 2030 Committee) Mar 2024 - Present

Electronics repair workshops for students

Skills developed: fault diagnosis, technical teaching & knowledge transfer, sustainability-oriented engineering practice.

Additional Projects 2025 - 2026

Skills developed: embedded video streaming (ESP32-S3, MJPEG/HTTP), computer vision & SLAM (feature matching, multiview geometry, transfer learning with PyTorch/Keras).

EDUCATION

B.Sc. Electronic and Automation Engineering (Robotics & Automation) 2022 - 2027 (exp.)
EINA, Universidad de Zaragoza, Spain · IEEE SB UZ, Agenda 2030 Committee

B.Sc. Mechanical Engineering - Exchange Year 2026 - 2027
Nanjing Tech University, China · Scholarship (academic record)

Blended Intensive Program - Global Value Chain in Bike Manufacturing Mar - Apr 2026
Hochschule Esslingen, Germany

Mandarin Chinese (HSK 2) 2025 - 2026
Confucius Institute, Zaragoza

English B2 2021 - 2022
Escuela Oficial de Idiomas, Zaragoza

ACHIEVEMENTS & OUTREACH

- **Winner**- 1st Prize · Best Performance Award · JRG-26 at the 10th ASTI Robotics Challenge (ARC 2026).
- Participant - NASA Space Apps 2025, WED 2026 Hackathon.
- Live robotics outreach stand at the XVII Engineering and Architecture Week (23-26 Mar 2026): adapted technical content (PID, odometry, wireless comms) for university students, schoolchildren and the public.